



SEMESTER I 2019/2020

SYSTEM ANALYSIS AND DESIGN (SCSD2613)

PHASE 1

PROJECT PROPOSAL & PLANNING REPORT

PREPARED BY:

- | | |
|--|-------------|
| 1. JUDITH ALOYSIUS SITIOLO | (A18CS0087) |
| 2. TAN CHONG LIM | (A18CS0255) |
| 3. NURFA HADIRAH BINTI TABRI | (A18CS0206) |
| 4. PUTERA MUHAMMAD SYABIL BIN SARIANTO | (A18CS0235) |

STAKEHOLDER: LH KERETA SEWA JB

PROJECT NAME: LH CAR RENTAL

SECTION: 04

LECTURER: DR AHMAD FADHIL YUSOF

TABLE OF CONTENTS

1.0 INTRODUCTION	3
2.0 BACKGROUND STUDY	3
3.0 PROBLEM STATEMENT	4
4.0 PROPOSED SOLUTION	5
4.1 FEASIBILITY STUDY	6
4.1.1 Technical Feasibility	6
4.1.2 Operational Feasibility	6
4.1.3 Economic Feasibility	7
5.0 OBJECTIVES	9
6.0 SCOPE	9
6.1 SYSTEM	9
6.2 USER	9
6.3 FEASIBILITY STUDY	10
7.0 PROJECT PLANNING	11
7.1 Human Resource	11
7.2 Work Breakdown Structure (WBS)	12
7.3 PERT Chart	12
7.4 Gantt Chart	13
8.0 BENEFIT AND SUMMARY OF PROPOSED SYSTEM	14
Appendix	15

1.0 INTRODUCTION

We are a group of undergraduate Software Engineering students from University Teknologi Malaysia (UTM). As in the requirement of our subject, System Analysis & Design (SCSD2613), we need to conduct a project for our assessment under this course. We have decided to propose a solution for a company to improve their management on the business and at the same time solve the problems they encounter during their business.

2.0 BACKGROUND STUDY

LH Kereta Sewa JB is a company located in Johor Bahru, Johor. It has over 5 years of experience in car rental field. The company provides car rental service for customers who need a ride especially at Johor area. There are several car models available for the customer to choose, including Viva, Axia, Myvi, etc. There are also choices between auto or manual cars available. The rental prices vary based on the time, date, and model of car to be rented.

Currently, all the bookings are done through 'Whatsapp' or by calling on mobile. When the customer wants to rent a car, they need to contact the company for car availability, price and time to take the car. After providing the required documents such as softcopy of IC and license through 'Whatsapp', they need to make arrangement to take the car and paying the fees. If the customer wants to extend the car rental time, they need to contact again the company for availability and add extra fees for that. All the steps are done manually, including the calculation of monthly profit margin.

3.0 PROBLEM STATEMENT

1. Answering repeating questions

Customers always asking repeating questions such as car price, car availability, document needed for car rental.

2. Manual arrangement of car rental booking schedule

Need to arrange for car rental schedule manually, and check for car availability manually every time customer want to book a car. When customer want to extend the renting time, they need to inform the company manually via phone call or Whatsapp.

3. Customer cannot find the location to take the car

Sometimes the customers get tend to get lost on their way to the location, although the location already provided. They do not know how to use the navigation app such as google map.

4. Calculating the profit margin

Since the company does not have any online/offline system, they need to key in the data monthly and calculate their expenses and profit manually by using Microsoft Excel. All calculation of profit margin needs to be done manually.

5. Manual update of price and announcement

Any changes of price of rental need to be updated manually and tell the customer every time. There are also certain conditions need to be satisfied such as minimum 5 days rental time during holiday seasons. Some customers might not know these conditions.

6. Communication problem

Some customer failed to inform the company that they have returned the car. The company need to remind and update the customer about the time for return the car. It is trouble for other customers who are waiting for the car.

4.0 PROPOSED SOLUTION

LH CAR RENTAL is a mobile application that allows the user to experience a new way of renting a car. This application will provide some features to overcome problems that users are facing.

With this application, user can create an account anywhere and anytime. It only needs one-way interaction. The user can key in their information such as name, age, address and email. The user also needs to snap a picture of their license and identity card and upload it during the registration. A verification email will be sent to the user's email account and the user needs to activate their account. After the registration, the user can log in to their account and use the application.

The next features, the user can book and rent the car anytime. The user can choose which car they want to rent, set time and place to pick up. There is a build-in navigation function in the app to navigate the customer to the correct location. The application will take the users information and reserve the car for the user. The application will proceed to the payment if the user is confirmed with their choice. The user can only pay using online banking whenever using this application.

Instead of paying extra to the company if the user sends the car late. This application allows the user to add some hours of the renting period. If the user wishes to add more time for renting period, the user can use this feature and it will lead to the payment page.

This app will automatically generate monthly revenue as per request by the company. The company no longer needed to calculate the profit manually.

4.1 FEASIBILITY STUDY

4.1.1 Technical Feasibility

This system uses mobile application as its platform, which only required a smartphone and internet access to use this app. This system needs a server and database system since the application needs to store the user's information and pictures in order to create an account. The system also needs to have a strong security since it uses online payment to rent a car. Hence, it is technically feasible.

4.1.2 Operational Feasibility

The mobile application has a database and online payment feature. This means that the system requires an expert like IS support to supervise and maintain the operation of the system, database and security. This can aid the system if there's a bug or glitch. IS support can also help the company to do advertisement like make a new home screen for a promotion week. The owner of the company can manage and track the car rental information while the users can only view and book for car.

4.1.3 Economic Feasibility

Cost-Benefits Analysis

Estimated Cost	
Hardware	RM25,000
Software	RM20,000
Advertisement	RM3,000 per year
Maintenance	RM8,000 per year
Transport	RM1,000 per year
IS support	RM3,000 per year

Estimated Benefits	
Increase sales	RM20,000
Saving	RM15,000

Assumption	
Discount rate	10%
Sensitivity factor (cost)	1.1
Sensitivity factor(benefits)	1.3
Annual change in production cost	4%
Annual change in benefits	7%

Costs	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Development Costs						
Hardware	27500					
Software	22000					
Total	49500					
Production Cost						
Advertisement		3300	3432	3569	3711	3859
Maintenance		8800	9152	9518	9899	10294
Transport		1100	1144	1190	1238	1288
IS support		3300	3432	3569	3711	3859
Annual Production Cost		16500	17160	17847	18559	19300
Present Value		15000	14182	13409	12676	11984
Accumulated Cost		64500	78682	92091	104767	116751

Benefits	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Increase sales		26000	27820	29767	31851	34081
Saving		19500	20865	22326	23889	25561
Annual benefits		45500	48685	52093	55740	59642
Present Value		41364	40236	39138	38071	37033
Accumulated Benefits		41364	81600	120738	158809	195842
Gain or Loss		-23136	2918	28647	54042	79091
Profitability Index	1.59					

From the table above, we can see that the profitability index is 1.59. The index is a good investment because the value is greater than 1. Even though we lose around RM23,136 in the first year, it increases gradually after that. LH CAR RENTAL mobile application is estimated to gain RM79,091. Thus this is a good project with a return.

5.0 OBJECTIVES

- I. To design a mobile application for the car rental company to ease the renting process for the users.
- II. To create a system that provides information such as the type of cars that available for renting.
- III. To provide more secure procedure for online payment, pick-up and returning car keys.
- IV. To ease user's data collection such as IC and driving license for the car rental company.
- V. To ease calculation of monthly expenses and profit.

6.0 SCOPE

6.1 SYSTEM

MOBILE APPLICATION

- Allow the user to view information of car rental
- Allow the user to create an account and login
- Provide a secure and accurate data collection

6.2 USER

CUSTOMER

- Allow user to view information
- Allow user to insert name, age, address and e-mail
- Allow user to add more rental hours if needed
- User login
- Allow user to inquire the details of the car.
- Notification system

6.3 FEASIBILITY STUDY

1. TECHNICAL

- Mobile application as main platform
- Use database system to store information
- Security measurements for online payment

2. OPERATIONAL

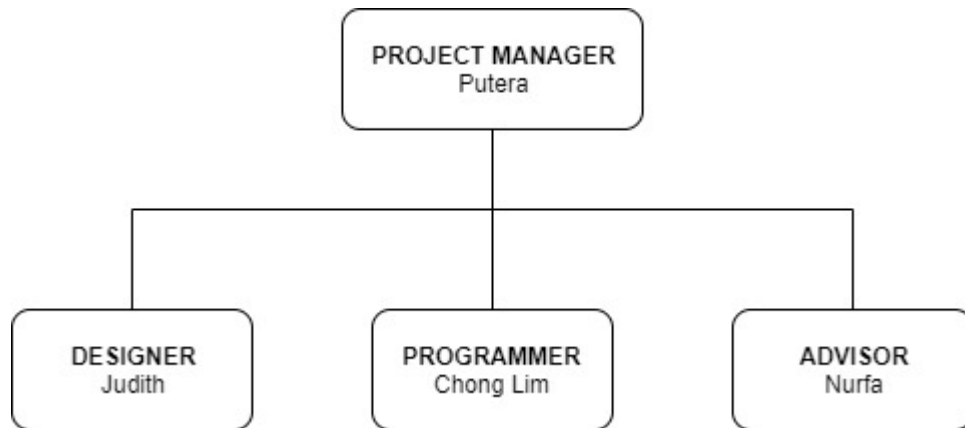
- Advertising (make a promotion)
- Human resource (management, budget and timeline)
- Efficiency
- Maintenance

3. ECONOMIC

- Hardware and software cost
- Testing cost
- Application cost
- Duration to design and build an application

7.0 PROJECT PLANNING

7.1 Human Resource



Project Manager: Putera Muhammad Syabil Bin Sarianto

- Manages tasks and progress of such tasks
- Assigns tasks to people

Designer: Judith Aloysius Sitiol

- Prepare rough design for website wireframe
- Refine design for website prototype

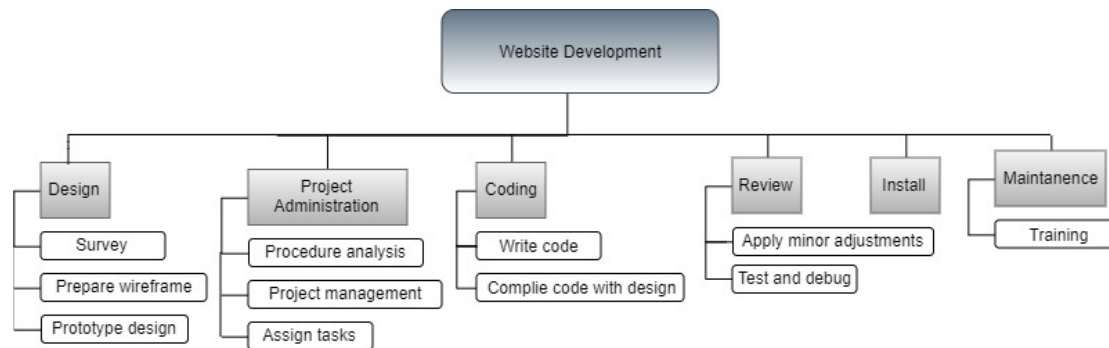
Programmer: Tan Chong Lim

- Build a programmable website based on design

Advisor: Nurfa Hadirah Binti Tabri

- Reviews progress
- Acknowledge and points out parts that can be improved

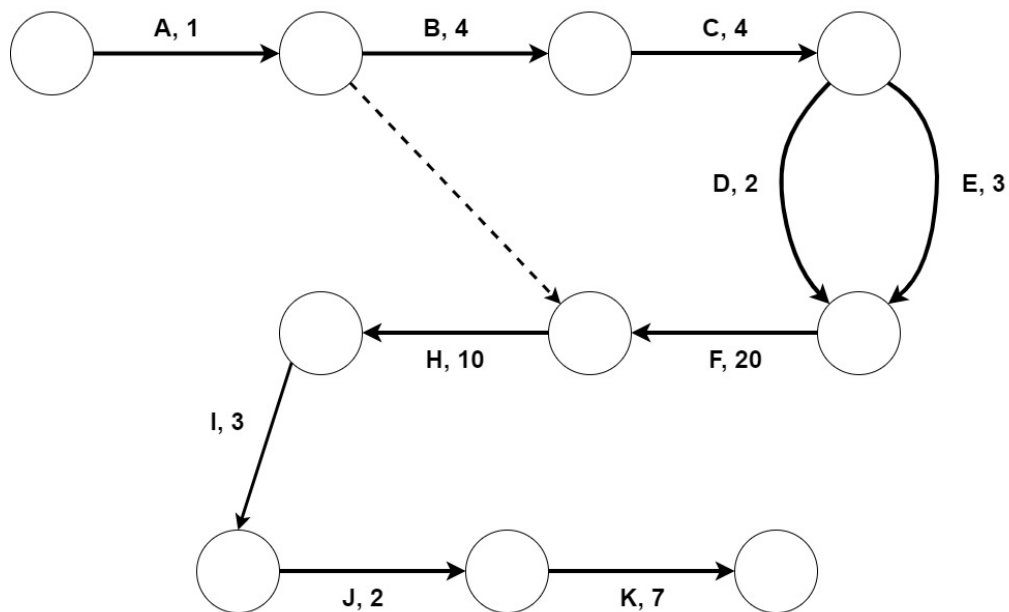
7.2 Work Breakdown Structure (WBS)



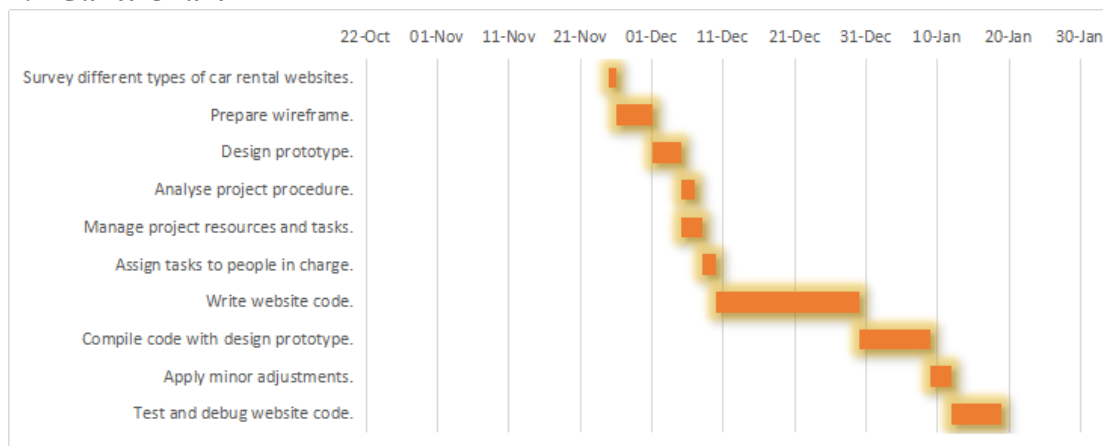
7.3 PERT Chart

Activity	Description	Predecessor	Expected Time
A	Survey different types of car rental websites.	None	1 Day (25/11/19)
B	Prepare wireframe.	A	5 Days (26/11/19 - 30/11/19)
C	Design prototype.	B	4 Days (1/12/19 - 4/12/19)
D	Analyse project procedure.	C	2 Days (5/12/19 - 6/12/19)
E	Manage project resources and tasks.	C	3 Days (5/12/19 - 7/12/19)
F	Assign tasks to people in charge.	D, E	2 Days (8/12/19 - 9/12/19)
G	Write application code.	F	20 Days (10/12/19 - 29/12/19)
H	Compile code with design prototype.	B, G	10 Days (30/12/19 - 8/1/20)
I	Apply minor adjustments.	H	3 Days (9/1/20 - 11/1/20)
J	Test and debug website code.	I	7 Days

			(12/1/20 - 18/1/20)
K	Install software for application management.	J	2 Days (19/1/20 - 20/1/20)
L	Provide training for new employees.	K	7 Days (21/1/20 - 27/1/20)



7.4 Gantt Chart



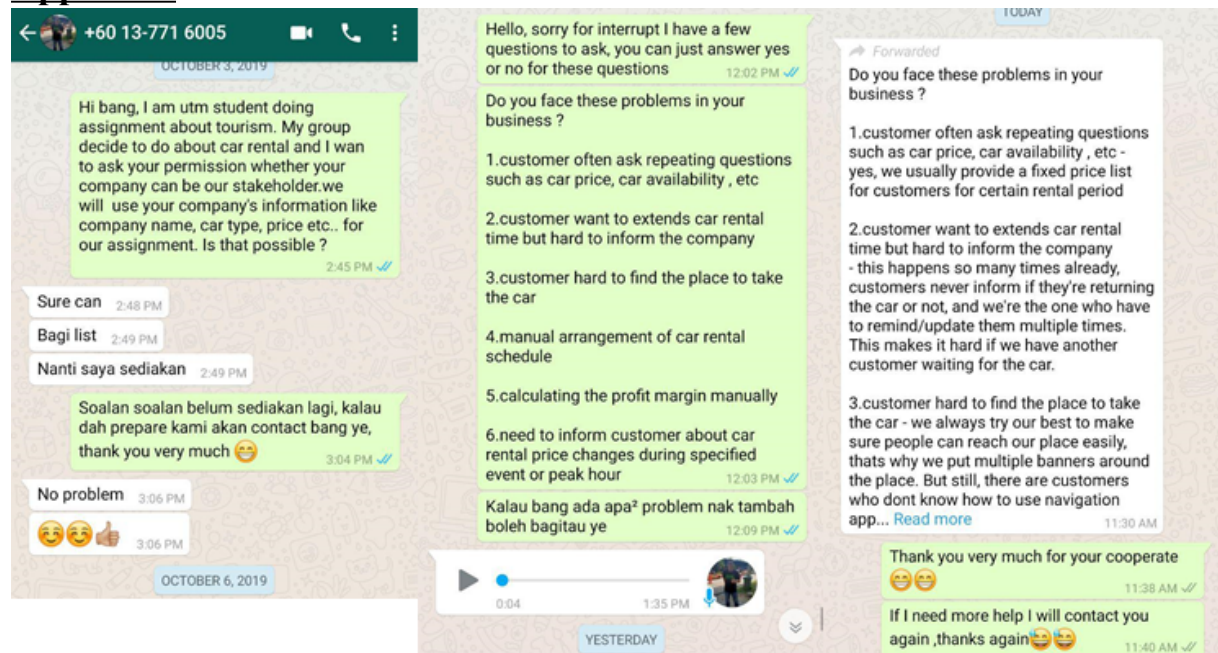
8.0 BENEFIT AND SUMMARY OF PROPOSED SYSTEM

This app will benefit both the users and the company. For the company, they can save their time for calculation of monthly profit because it is all auto generated by the app. Besides, the company no need repeat answering the same questions such as price, car availability as it is all shown in the app clearly. All announcement or changes of price can be made in the app and everyone with the app will be updated instantaneously.

For the customer, they can obtain all information from the app, without need to wait for the company to reply them via Whatsapp. They can just book for an available car from their phone, with just a few clicks. Besides, they will be navigated to the right location by using the app, without needed to installing others app. The users will also be notified when their rental time is over. It acts as a reminder for them.

It can be said that this app save time for both the parties and makes the booking steps easier.

Appendix



Appendix 1: Conversation with stakeholder